

CBSE Class 10 Mathematics (Standard) 2020

Board Exam Previous Year Paper | Complete Solutions

Exam Date: 12 March 2020 | **Time:** 3 Hours | **Max Marks:** 80

Total Questions: 40, divided into 4 sections — A, B, C, D

Section A: Q1–20, 1 Mark each (20) | **Section B:** Q21–26, 2 Marks each (12)

Section C: Q27–34, 3 Marks each (24) | **Section D:** Q35–40, 4 Marks each (24)

General Instructions: All questions are compulsory. Internal choice is provided in a few questions in each section. Use of calculators is NOT allowed. Draw neat diagrams wherever required.

Yeh CBSE Class 10 Maths (Standard) 2020 ka asli board exam paper hai — lockdown se pehle, 12 March 2020 ko conduct hua tha. Poora syllabus wala 80-marks single paper (Term system nahi).

SECTION A — Q1 to Q20 (1 Mark each)

Q1. The HCF of 96 and 404 is 4. What is their LCM? [1 Mark]

■ ANSWER / UTTAR

$$\text{LCM} = (96 \times 404) / \text{HCF} = (96 \times 404) / 4 = 9696.$$

Q2. Is $\sqrt{5}$ a rational or irrational number? [1 Mark]

■ ANSWER / UTTAR

$\sqrt{5}$ is an **irrational number** — it cannot be expressed as p/q ($q \neq 0$).

Q3. If one zero of the polynomial $p(x) = x^2 - 5x + k$ is 2, find k . [1 Mark]

■ ANSWER / UTTAR

$$p(2)=0 \Rightarrow 4-10+k=0 \Rightarrow k = 6.$$

Q4. For what value of k do the equations $2x+3y=5$ and $4x+ky=10$ represent coincident lines? [1 Mark]

■ ANSWER / UTTAR

$$\text{Ratios equal: } 2/4 = 3/k = 5/10 \Rightarrow 3/k = 1/2 \Rightarrow k = 6.$$

Q5. Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$. [1 Mark]

■ ANSWER / UTTAR

$$D = b^2 - 4ac = 16 - 24 = -8 \text{ (no real roots).}$$

Q6. Find the 10th term of the AP: 2, 7, 12, 17, ... [1 Mark]

■ ANSWER / UTTAR

$$a=2, d=5. a_{10} = a+9d = 2+45 = 47.$$

Q7. Find the distance between points A(2,3) and B(4,1). [1 Mark]

■ ANSWER / UTTAR

$$d = \sqrt{[(4-2)^2 + (1-3)^2]} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \text{ units.}$$

Q8. Find the midpoint of the line segment joining (-3,5) and (7,1). [1 Mark]

■ ANSWER / UTTAR

$$\text{Midpoint} = ((-3+7)/2, (5+1)/2) = (2, 3).$$

Q9. In $\triangle ABC$, $DE \parallel BC$. If $AD=2$, $DB=3$, $AE=4$, find EC . [1 Mark]

■ ANSWER / UTTAR

$$\text{By Basic Proportionality Theorem: } AD/DB = AE/EC \Rightarrow 2/3 = 4/EC \Rightarrow EC = 6.$$

Q10. Two concentric circles have radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle. [1 Mark]

■ ANSWER / UTTAR

$$\text{Half-chord}^2 = 5^2 - 3^2 = 16, \text{ half-chord} = 4, \text{ chord} = 8 \text{ cm.}$$

Q11. Find the value of $\sin 30^\circ + \cos 60^\circ$. [1 Mark]

■ ANSWER / UTTAR

$$1/2 + 1/2 = 1.$$

Q12. Simplify: $(1 - \cos^2\theta)/\sin^2\theta$. [1 Mark]

■ ANSWER / UTTAR

$$= \sin^2\theta/\sin^2\theta = 1.$$

Q13. If $\tan\theta = 1$, find the value of θ ($0^\circ \leq \theta \leq 90^\circ$). [1 Mark]

■ ANSWER / UTTAR

$$\tan 45^\circ = 1, \text{ so } \theta = 45^\circ.$$

Q14. Find the area of a circle of radius 7 cm. ($\pi = 22/7$) [1 Mark]

■ ANSWER / UTTAR

$$\text{Area} = \pi r^2 = 22/7 \times 49 = 154 \text{ cm}^2.$$

Q15. A cone has radius 3.5 cm and height 12 cm. Find its volume. ($\pi = 22/7$) [1 Mark]

■ ANSWER / UTTAR

$$V = (1/3)\pi r^2 h = (1/3) \times 22/7 \times 12.25 \times 12 = 154 \text{ cm}^3.$$

Q16. Find the mean of the first 5 natural numbers. [1 Mark]

■ ANSWER / UTTAR

$$(1+2+3+4+5)/5 = 15/5 = 3.$$

Q17. Define the median of grouped data in one line. [1 Mark]

■ ANSWER / UTTAR

The median is the value that **divides the data into two equal halves** when arranged in order, found using the cumulative frequency curve/formula for grouped data.

Q18. A die is thrown once. Find the probability of getting a prime number. [1 Mark]

■ ANSWER / UTTAR

$$\text{Primes on a die: } 2, 3, 5 \Rightarrow P = 3/6 = 1/2.$$

Q19. Two coins are tossed together. Find the probability of getting exactly one head. [1 Mark]

■ ANSWER / UTTAR

$$\text{Outcomes: HH, HT, TH, TT. Favourable: HT, TH} \Rightarrow P = 2/4 = 1/2.$$

Q20. What is the maximum number of tangents that can be drawn from an external point to a circle? [1 Mark]

■ ANSWER / UTTAR

Two tangents can be drawn from an external point to a circle.

SECTION B — Q21 to Q26 (2 Marks each)

Q21. Find the HCF and LCM of 26 and 91, and verify that $\text{HCF} \times \text{LCM} = \text{product of the two numbers}$. **[2 Marks]**

■ ANSWER / UTTAR

$26 = 2 \times 13$, $91 = 7 \times 13$. $\text{HCF} = 13$, $\text{LCM} = (2 \times 7 \times 13) = 182$.

Verify: $13 \times 182 = 2366 = 26 \times 91 = \mathbf{2366}$ ✓.

Q22. Find the zeroes of the quadratic polynomial $x^2 - 3x - 10$ and verify the relationship between the zeroes and coefficients. **[2 Marks]**

■ ANSWER / UTTAR

$x^2 - 3x - 10 = (x - 5)(x + 2) \Rightarrow \text{zeroes} = 5, -2$.

Sum = $5 + (-2) = 3 = -(-3)/1$ ✓; Product = $-10 = -10/1$ ✓.

Q23. Solve for x and y: $x + y = 10$, $x - y = 4$. **[2 Marks]**

■ ANSWER / UTTAR

Adding: $2x = 14 \Rightarrow x = 7$. Then $y = 10 - 7 = 3$. So $x = 7$, $y = 3$.

Q24. Which term of the AP 21, 18, 15, ... is -81? Also find the term which is zero. **[2 Marks]**

■ ANSWER / UTTAR

$a = 21$, $d = -3$. $a + (n - 1)(-3) = -81 \Rightarrow (n - 1) = 34 \Rightarrow n = \mathbf{35\text{th term}}$.

For zero: $21 - 3(n - 1) = 0 \Rightarrow n - 1 = 7 \Rightarrow n = \mathbf{8\text{th term is 0}}$.

Q25. Prove: $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2 = 2$. **[2 Marks]**

■ ANSWER / UTTAR

Expand: $(\sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta) + (\sin^2\theta - 2\sin\theta\cos\theta + \cos^2\theta) = 2\sin^2\theta + 2\cos^2\theta = 2(\sin^2\theta + \cos^2\theta) = \mathbf{2}$ ✓.

Q26. A bag contains 5 red and 7 blue balls. One ball is drawn at random. Find the probability that it is (i) red (ii) not blue. **[2 Marks]**

■ ANSWER / UTTAR

Total balls = 12.

(i) $P(\text{red}) = 5/12$

(ii) 'Not blue' = red, so $P(\text{not blue}) = \mathbf{5/12}$.

SECTION C — Q27 to Q34 (3 Marks each)

Q27. The sum of the ages of a father and his son is 45 years. Five years ago, the product of their ages was 124. Find their present ages. [3 Marks]

■ ANSWER / UTTAR

Let son's present age = x , father's = $45-x$.

5 yrs ago: $(x-5)(45-x-5)=124 \Rightarrow (x-5)(40-x)=124$

$40x - x^2 - 200 + 5x = 124 \Rightarrow -x^2 + 45x - 324 = 0 \Rightarrow x^2 - 45x + 324 = 0$

$x = [45 \pm \sqrt{(2025-1296)}]/2 = [45 \pm 27]/2 \Rightarrow x=36 \text{ or } x=9$

Since father's age must be reasonable, **son = 9 years, father = 36 years.**

Q28. Find the ratio in which the point $(-4, 6)$ divides the line segment joining $A(-6,10)$ and $B(3,-8)$. [3 Marks]

■ ANSWER / UTTAR

Let ratio = $k:1$. Using section formula on x : $-4 = (3k-6)/(k+1) \Rightarrow -4k-4=3k-6 \Rightarrow 2=7k \Rightarrow k=2/7$.

So the ratio is **2:7**.

Q29. In $\triangle ABC$, if a line DE parallel to BC divides AB and AC such that $AD=4\text{cm}$, $DB=4.5\text{cm}$, $AE=8\text{cm}$, find EC . [3 Marks]

■ ANSWER / UTTAR

By Basic Proportionality Theorem: $AD/DB = AE/EC \Rightarrow 4/4.5 = 8/EC \Rightarrow EC = (8 \times 4.5)/4 = \mathbf{9 \text{ cm}}$.

Q30. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact (statement + reasoning outline). [3 Marks]

■ ANSWER / UTTAR

Let O be the centre, P the point of contact, and XY the tangent at P . Take any other point Q on XY and join OQ . Since Q lies outside the circle, $OQ > OP$ (OP is the radius, the shortest distance from O to line XY).

This is true for every point on XY except P , which means **OP is the shortest distance from O to XY , so $OP \perp XY$** — i.e., the tangent is perpendicular to the radius at the point of contact.

Q31. From a point on the ground, the angle of elevation of the top of a 20 m tall tower is 30° . Find the distance of the point from the base of the tower. [3 Marks]

■ ANSWER / UTTAR

$\tan 30^\circ = 20/d \Rightarrow 1/\sqrt{3} = 20/d \Rightarrow d = 20\sqrt{3} \approx \mathbf{34.64 \text{ m}}$.

Q32. Find the area of a sector of a circle with radius 21 cm and central angle 60° . ($\pi=22/7$) [3 Marks]

■ ANSWER / UTTAR

Area = $(\theta/360) \times \pi r^2 = (60/360) \times 22/7 \times 441 = (1/6) \times 1386 = \mathbf{231 \text{ cm}^2}$.

Q33. Find the mean of the following data using the direct method:

Class: 0-10, 10-20, 20-30, 30-40; Frequency: 4, 6, 8, 2 [3 Marks]

■ ANSWER / UTTAR

Midpoints (x): 5, 15, 25, 35. $f \cdot x$: 20, 90, 200, 70. $\Sigma f = 20$, $\Sigma f \cdot x = 380$.

Mean = $380/20 = \mathbf{19}$.

Q34. Find the sum of the first 24 terms of the AP whose n th term is $a_n = 3 + 2n$. [3 Marks]

■ ANSWER / UTTAR

$a_1=5$, $a_{24}=7$, so $a=5$, $d=2$. $S_{24} = (24/2)[2(5)+(23)(2)] = 12[10+46] = 12 \times 56 = \mathbf{672}$.

SECTION D — Q35 to Q40 (4 Marks each)

Q35. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the original speed of the train. [4 Marks]

■ ANSWER / UTTAR

Let speed = x km/h. Time = $360/x$. New: $360/(x+5) = 360/x - 1$.

$$360x - 360(x+5) = -x(x+5) \Rightarrow -1800 = -x^2 - 5x \Rightarrow x^2 + 5x - 1800 = 0$$

$$x = \frac{-5 \pm \sqrt{(25+7200)}}{2} = \frac{-5 \pm 85}{2} \Rightarrow x=40 \text{ (rejecting negative)}$$

Original speed = 40 km/h.

Q36. The sum of the first n terms of an AP is given by $S_n = 3n^2 + 4n$. Find the AP and its 15th term. [4 Marks]

■ ANSWER / UTTAR

$$a_1 = S_1 = 3 + 4 = 7.$$

$$S_2 = 12 + 8 = 20 \Rightarrow a_2 = S_2 - S_1 = 13.$$

$$d = a_2 - a_1 = 6. \text{ AP: } 7, 13, 19, 25, \dots$$

$$a_{15} = a_1 + 14d = 7 + 84 = 91.$$

Q37. State and prove the Basic Proportionality Theorem (Thales' Theorem): If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. [4 Marks]

■ ANSWER / UTTAR

Given: $\triangle ABC$ with $DE \parallel BC$, D on AB , E on AC .

To Prove: $AD/DB = AE/EC$.

Construction: Join BE , CD . Draw $DM \perp AC$ and $EN \perp AB$.

Proof: $\text{ar}(\triangle ADE)/\text{ar}(\triangle DBE) = AD/DB$ (same height EN , bases AD & DB)

$\text{ar}(\triangle ADE)/\text{ar}(\triangle DEC) = AE/EC$ (same height DM , bases AE & EC)

Since $\triangle DBE$ and $\triangle DEC$ lie between the same parallels DE and BC and on the same base DE ,

$$\text{ar}(\triangle DBE) = \text{ar}(\triangle DEC).$$

Therefore, $AD/DB = AE/EC$. **Hence proved.**

Q38. Prove that the lengths of two tangents drawn from an external point to a circle are equal. [4 Marks]

■ ANSWER / UTTAR

Given: A circle with centre O , and two tangents PA and PB drawn from an external point P , touching the circle at A and B .

To Prove: $PA = PB$.

Construction: Join OA , OB and OP .

Proof: Since PA and PB are tangents, $OA \perp PA$ and $OB \perp PB$ (radius $\perp$ tangent). In $\triangle OAP$ and $\triangle OBP$:

$OA = OB$ (radii), $OP = OP$ (common), $\angle OAP = \angle OBP = 90^\circ$.

By RHS congruence, $\triangle OAP \cong \triangle OBP$.

Therefore, by CPCT, **$PA = PB$. Hence proved.**

Q39. The angle of elevation of the top of a tower from a point on the ground is 30° . On walking 40 m towards the tower, the angle of elevation becomes 60° . Find the height of the tower. [4 Marks]

■ ANSWER / UTTAR

Let height = h , and distance from tower after walking = x .

$$\tan 60^\circ = h/x \Rightarrow x = h/\sqrt{3}$$

$$\tan 30^\circ = h/(x+40) \Rightarrow 1/\sqrt{3} = h/(x+40) \Rightarrow x+40 = h\sqrt{3}$$

$$\text{Substituting: } h/\sqrt{3} + 40 = h\sqrt{3} \Rightarrow 40 = h\sqrt{3} - h/\sqrt{3} = h(3-1)/\sqrt{3} = 2h/\sqrt{3}$$

$$h = 40\sqrt{3}/2 = 20\sqrt{3} \approx \mathbf{34.64 \text{ m.}}$$

Q40. A solid is in the form of a cylinder with hemispherical ends. The total height of the solid is 19 cm and the radius of the cylinder (and hemispheres) is 3.5 cm. Find the total surface area of the solid.

($\pi=22/7$) **[4 Marks]**

■ **ANSWER / UTTAR**

Radius $r=3.5$ cm. Height of cylindrical part = $19 - 2(3.5) = 12$ cm.

TSA = CSA of cylinder + CSA of 2 hemispheres

$$= 2\pi rh + 2(2\pi r^2) = 2\pi r(h+2r)$$

$$= 2 \times 22/7 \times 3.5 \times (12+7) = 22 \times 19 = \mathbf{418 \text{ cm}^2}.$$

■ **Key Takeaways:** Section A ke MCQ/VSA questions speed-solve karo — zyada time mat lagao. Section D ke proofs (Q37, Q38) word-for-word structure yaad rakho — CBSE step-marking deti hai.